

WGS Unknown NTM_22-11 Investigation

Preface

We have recieved unknown samples of Non-tuberculous mycobacteria (NTM) collected in Hawai'i, a region with the highest reported incidence of NTM infection in the United States. NTMs are a diverse group of over 200 speices of bacteria found in soil,freshwater,facuets and showerheads, where they can multiply. Exposure to these types of environments, particularly in Hawaii's humid environment, promotes NTM survival and transimission, posing as a huge risk for the public. The National Institute of Health stated that Hawaii has two-fold greater than the national average risk of infection, causing issues in the lungs. Despite the dangers of NTMS, the diversity of NTMs in Hawai'i is still incompletely discovered. A clinical laboratory in Hawaii collected NTM_22-11 as a potentially novel NTM. In order to confirm the identity of the samples, we have performed whole genome sequencing (WGS) on the samples using the Oxford Nanopore MinION sequencing platform. Samples were prepared using the rapid barcoding kit (SQK-RBK114.24) and sequenced on a MinION flow cell. Raw sequencing data were basecalled and demultiplexed with Doroado v0.8.2, where reads were filtered using Chopper. Draft genomes were assembled using Flye v2.8.3 and polished using Medaka v1.0.3. Genomic characterization of NTM_22-11 will help us discover potentially novel speicies that affect the public health.

```
library(stringr)
library(dplyr)
library(ggplot2)
library(knitr)
library(kableExtra)
library(ape)
```

Completeness Check

Here, we want to measure the genome completeness. Using Docker and CheckM, it allows us to estimate how much of the expected bacterial genome is present by searching genes that should exist in all bacteria. Where the clincial labatory in Hawaii collected samples, there is a possibility of other environmental bacteria present in the data. CheckM flags contamination so that genome size isn't inflated. Overall Checkm is used to make sure NTM_22-11 is accurately represented as a single bacteria.

```
# Running checkM
# Place the docker command for checkM here
docker pull nanozoo/checkm
docker run --name=checkm \
--rm \
--volume=$PWD:/home \
--workdir=/home \
-it nanozoo/checkm:latest

checkm taxonomy_wf domain Bacteria -t 2 -x fasta ./ ./checkm_out

checkm qa -o 2 --tab_table -t 2 \
-f checkm_out/summary.tsv checkm_out/Bacteria.ms checkm_out
```

```

# Read in the checkM summary
checkm <- read.csv("checkm_out/summary.tsv", sep = "\t")

# Create tables for checkM completion and contamination
# Make sure to include the following fields: Bin.Id, Marker.lineage, Completeness, Contamination
# Make sure to rename the columns to ID, Lineage, Completeness, Contamination
checkm_table <- data.frame(checkm)
checkm_table_renamed <- checkm_table %>%
  select(ID = Bin.Id, Lineage = Marker.lineage, Completeness = Completeness, Contamination = Contamination)

# Create table for checkM additional summary stats
# Make sure to include the following fields: Bin.Id, Genome.size..bp., GC, X..contigs, N50..contigs.
# Make sure to rename the columns to ID, Genome Size (bp), GC, Contig #, N50
checkm_stats <- checkm_table %>%
  select(Bin.Id, Genome.size..bp., GC, X..contigs, N50..contigs.) %>%
  rename(ID = Bin.Id, Genome_Size_bp = Genome.size..bp., Contigs = X..contigs, N50 = N50..contigs.)

# Print tables
# checkm_table_renamed %>%
#   kable("latex", booktabs = TRUE, caption = "CheckM Overview",
#     col.names = "Overview stats") %>%
#   kable_styling(latex_options = c("striped", "hold_position"))

kable(checkm_table_renamed)

```

ID	Lineage	Completeness	Contamination
NTM_22-11	Bacteria	97.13	0

```

# checkm_stats %>%
#   kable("latex", booktabs = TRUE, caption = "CheckM Summary Stats",
#     col.names = "Summary stats") %>%
#   kable_styling(latex_options = c("striped", "hold_position"))

kable(checkm_stats)

```

ID	Genome_Size_bp	GC	Contigs	N50
NTM_22-11	4201960	66.7	1	4201960

Prokka Annotation

The next step is to turn raw DNA sequences into something that is readable. The genome contain the string of “A”, “T”, “C”, “G”. Prokka allows us to look at the tRNA, rRNA and other elements within the genome to conduct an analysis. Prokka also makes sure our NTM is novel since the gene content allows us to distinguish between different NTMs.

```

# Running Prokka
# Place the docker command for Prokka here
docker run --name=prokka \

```

```
--rm \
--volume=$PWD:/data \
-it staphb/prokka:latest

prokka --outdir prokka_out --prefix prokka_annotation \
--cpus 8 NTM_22-11.fasta
```

```
# Read in the prokka annotation
prokka <- read.csv("prokka_out/prokka_annotation.tsv", sep = "\t")
# Summarize the prokka annotation
# You will need to count the number of CDS, tRNA, rRNA, and non-hypothetical CDS
# Number of genes annotated
#prokka$ftype[1]
CDS_count <- 0
rRNA_count <- 0
tRNA_count <- 0
non_hypoCDS_count <- 0
for(i in 1:nrow(prokka)){
  if(prokka$ftype[i] == "CDS"){
    CDS_count <- CDS_count + 1
  }else if(prokka$ftype[i] == "rRNA"){
    rRNA_count <- rRNA_count + 1
  }else if(prokka$ftype[i] == "tRNA"){
    tRNA_count <- tRNA_count + 1
  }
}

for(i in 1:nrow(prokka)){
  if(prokka$product[i] != "hypothetical protein"){
    non_hypoCDS_count <- non_hypoCDS_count + 1
  }
}

CDS_count
```

```
## [1] 4098
```

```
# Combine the counts into a table
# Make sure to clear the rownames for clean formatting
prokka_summary <- data.frame(
  CDS = CDS_count,
  tRNA = tRNA_count,
  rRNA = rRNA_count,
  non_hypoCDS = non_hypoCDS_count
)
```

```
# Print table
kable(prokka_summary)
```

CDS	tRNA	rRNA	non_hypoCDS
4098	74	12	2240

```
# prokka_summary %>%
#   kable("latex", booktabs = TRUE, caption = "Prokka Annotation Summary") %>%
#   kable_styling(latex_options = c("striped", "hold_position"))
```

```
# Check to see if 16S is present & extract if present
#use ape
prokka <- read.FASTA("prokka_out/prokka_annotation.ffn")
# class(prokka)
# str(prokka)
# names(prokka)
RNA <- prokka[grep("16S ribosomal", names(prokka))]
write.FASTA(RNA, file = "16S_unkownNTM.fasta")
```

BLAST 16S rRNA Identification

The 16S rRNA gene is used for identifying bacterial because it is in almost all bacteria, has conserved regions that is unique to each bacteria. Running BLASTn with our bacteria determines whether NTM_22-11's genomic representation is close to other bacteria species. When a result shows a high similarity to other bacteria but no exact matches, it tells us that NTM_22-11 could be novel.

```
# Running BLAST
# Place the docker command for BLAST here
Docker build -t 16sblast:local .
```

```
docker run --name=blastn \
--rm \
--volume=$PWD:/home \
--workdir=/home \
-it 16sblast:local
```

```
# Read in the BLAST results
blast <- read.delim("blast_results.txt", header = FALSE)
# Make sure to remove columns 4, 5, and 9
# Make sure to parse species name from column V8 and add it as a new column
# Make sure to rename the columns to Sample ID, Identity (%), Length, E-value, Bit Score, Species
drops <- c(colnames(blast[4]), colnames(blast[5]), colnames(blast[9]))
blast <- blast[, !names(blast) %in% drops]
```

```
# Create table
kable(blast)
```

V1	V2	V3	V6	V7	V8
AOKDPPHH_903758	1543	0	2776	Stenotrophomonas forensis strain DFS-20110405 16S ribosomal RNA, complete sequence	
AOKDPPHH_903758	1536	0	2765	Stenotrophomonas maltophilia strain IAM 12423 16S ribosomal RNA, partial sequence	

V1	V2	V3	V6	V7	V8
AOKDPPHH_903748	1519	0	2747	Stenotrophomonas hibiscicola strain ATCC 19867 16S ribosomal RNA, partial sequence	
AOKDPPHH_983748	1517	0	2712	Stenotrophomonas beteli strain ATCC 19861 16S ribosomal RNA, partial sequence	
AOKDPPHH_903738	1500	0	2700	Stenotrophomonas maltophilia strain LMG 958 16S ribosomal RNA, partial sequence	
AOKDPPHH_903828	1543	0	2765	Stenotrophomonas forensis strain DFS-20110405 16S ribosomal RNA, complete sequence	
AOKDPPHH_903803	1536	0	2765	Stenotrophomonas maltophilia strain IAM 12423 16S ribosomal RNA, partial sequence	
AOKDPPHH_903803	1519	0	2736	Stenotrophomonas hibiscicola strain ATCC 19867 16S ribosomal RNA, partial sequence	
AOKDPPHH_903803	1500	0	2700	Stenotrophomonas maltophilia strain LMG 958 16S ribosomal RNA, partial sequence	
AOKDPPHH_983803	1517	0	2700	Stenotrophomonas beteli strain ATCC 19861 16S ribosomal RNA, partial sequence	
AOKDPPHH_904179	1543	0	2765	Stenotrophomonas forensis strain DFS-20110405 16S ribosomal RNA, complete sequence	
AOKDPPHH_904179	1536	0	2765	Stenotrophomonas maltophilia strain IAM 12423 16S ribosomal RNA, partial sequence	
AOKDPPHH_904173	1519	0	2736	Stenotrophomonas hibiscicola strain ATCC 19867 16S ribosomal RNA, partial sequence	
AOKDPPHH_904173	1500	0	2700	Stenotrophomonas maltophilia strain LMG 958 16S ribosomal RNA, partial sequence	
AOKDPPHH_984179	1517	0	2700	Stenotrophomonas beteli strain ATCC 19861 16S ribosomal RNA, partial sequence	
AOKDPPHH_984134	1498	0	2641	Stenotrophomonas forensis strain DFS-20110405 16S ribosomal RNA, complete sequence	
AOKDPPHH_984134	1492	0	2632	Stenotrophomonas maltophilia strain IAM 12423 16S ribosomal RNA, partial sequence	
AOKDPPHH_984134	1481	0	2625	Stenotrophomonas hibiscicola strain ATCC 19867 16S ribosomal RNA, partial sequence	
AOKDPPHH_984169	1481	0	2601	Stenotrophomonas geniculata ATCC 19374 = JCM 13324 16S ribosomal RNA, partial sequence	
AOKDPPHH_984169	1473	0	2599	Stenotrophomonas pavanii strain LMG 25348 16S ribosomal RNA, partial sequence	

```
# blast %>%
#   kable("latex", booktabs = TRUE, caption = "16S BLAST Results",
#         col.names = c("Sample ID", "Identity (%)", "Length", "E-value", "Bit Score", "Species"))
```

Phylogenetic ID based on 16S rRNA

Next, we map the similar bacteria on a Phylogenetic tree to understand the evolution of each bacteria, and how close it is to NTM_22-11. The tree will visually tell us whether NTM_22-11 groups with other bacterial species or formed a separate branch from an existing bacteria. Using MUSCLE in this process will allow us to align the 16S rRNA gene to create the phylogenetic tree. IQ TREE allows us to test multiple models for the best fitted model when it comes to creating the phylogenetic tree. BLAST on its own isn't the most accurate way to tell if NTM_22-11 is novel since many bacterial species sharing almost the same 16S identity. The phylogenetic tree gives us an representation of the evolution of bacteria that can be similar to NTM_22-11.

```

one <- read.FASTA("16S_unkownNTM.fasta")
two <- read.FASTA("modified_representative_seqs.fasta")
three <- read.FASTA("outgroup.fasta")

out <- c(one, two, three)
write.FASTA(out, "combined_1.fasta")

```

```

# Running MUSCLE and IQ-TREE
# Place the docker command for MUSCLE and IQ-TREE here
#muscle
docker run --name=muscle \
  --rm \
  -v $PWD:/data \
  --workdir=/data \
  pegi3s/muscle -in /data/combined_1.fasta -out /data/musclecombined.fasta

#tree
docker run --name=iqtree2 \
  --rm \
  --volume=$PWD:/home \
  --workdir=/home \
  staphb/iqtree2:latest iqtree2 -s musclecombined.fasta -m TEST -B 1000 -T 8

```

```

# Read in the tree and root at outgroup
tree <- read.tree("musclecombined.fasta.contree")
# Use ?root for info
tree$tip.label

```

```

## [1] "AOKDPPHH_04184"
## [2] "AOKDPPHH_03798"
## [3] "AOKDPPHH_03803"
## [4] "AOKDPPHH_04179"
## [5] "NR_044590.1_Hoyosella_altamirensis"
## [6] "NR_158076.1_Mycobacterium_aquiterrae"
## [7] "NR_179750.1_Mycobacterium_neglectum"
## [8] "NR_180098.1_Mycobacterium_kyogaense"
## [9] "NR_119148.1_Mycobacteroides_abscessus"
## [10] "NR_119149.1_Mycobacteroides_chelonae"
## [11] "NR_117923.1_Mycobacteroides_franklinii"
## [12] "NR_043989.1_Mycobacteroides_salmoniphilum"
## [13] "NR_157700.1_Mycobacterium_stephanolepidis"
## [14] "NR_158073.1_Mycobacterium_chelonae"
## [15] "NR_145859.1_Mycobacteroides_saopaulense"
## [16] "NR_104876.1_Mycobacteroides_immunogenum"
## [17] "NR_115184.1_Mycobacterium_hackensackense"
## [18] "NR_178332.1_Mycobacterium_dioxanotrophicus"
## [19] "NR_158005.1_Mycobacterium_aquaticum"
## [20] "NR_179783.1_Mycobacterium_syngnathidarum"
## [21] "NR_115330.1_Mycobacterium_barrassiae"
## [22] "NR_159197.1_Mycobacterium_neumannii"
## [23] "NR_159196.1_Mycobacterium_lehmannii"
## [24] "NR_116504.1_Mycobacterium_paraterrae"
## [25] "NR_157699.1_Mycobacterium_talmoniae"

```

[26] "NR_157985.1_Mycobacterium_eburneum"
 ## [27] "NR_119156.1_Mycobacterium_shimoidei"
 ## [28] "NR_118882.1_Mycobacterium_xenopi"
 ## [29] "NR_028878.1_Mycobacterium_botniense"
 ## [30] "NR_028759.1_Mycobacterium_heckeshornense"
 ## [31] "NR_044442.1_Mycobacterium_noviomagense"
 ## [32] "NR_114661.1_Mycobacterium_cookii"
 ## [33] "NR_118426.1_Mycobacterium_fragae"
 ## [34] "NR_041897.1_Mycobacterium_branderi"
 ## [35] "NR_041663.1_Mycobacterium_kyorinense"
 ## [36] "NR_114894.1_Mycobacterium_celatum"
 ## [37] "NR_118917.1_Mycobacterium_asiaticum"
 ## [38] "NR_118331.1_Mycobacterium_gordonae"
 ## [39] "NR_178881.1_Mycobacterium_vicinigordonae"
 ## [40] "NR_125667.1_Mycobacterium_paragordonae"
 ## [41] "NR_147779.2_Mycobacterium_alsense"
 ## [42] "NR_178331.1_Mycobacterium_lepromatosis"
 ## [43] "NR_104888.1_Mycobacterium_haemophilum"
 ## [44] "NR_112623.1_Mycobacterium_shinjukuense"
 ## [45] "NR_113136.1_Mycobacterium_pseudoshottsii"
 ## [46] "NR_113138.1_Mycobacterium_ulcerans"
 ## [47] "NR_114890.1_Mycobacterium_marinum"
 ## [48] "NR_025636.1_Mycobacterium_shottsii"
 ## [49] "NR_178462.1_Mycobacterium_spongiae"
 ## [50] "NR_044826.2_Mycobacterium_tuberculosis"
 ## [51] "NR_178632.1_Mycobacterium_decipiens"
 ## [52] "NR_025175.1_Mycobacterium_lacus"
 ## [53] "NR_117225.1_Mycobacterium_malmoense"
 ## [54] "NR_181979.1_Mycobacterium_simulans"
 ## [55] "NR_044449.1_Mycobacterium_riyadhense"
 ## [56] "NR_145908.1_Mycobacterium_angelicum"
 ## [57] "NR_118584.1_Mycobacterium_szulgai"
 ## [58] "NR_026054.1_Mycobacterium_bohemicum"
 ## [59] "NR_025767.1_Mycobacterium_nebraskense"
 ## [60] "NR_115815.1_Mycobacterium_paraseoulense"
 ## [61] "NR_043904.1_Mycobacterium_seoulense"
 ## [62] "NR_119193.1_Mycobacterium_paraaffinicum"
 ## [63] "NR_118880.1_Mycobacterium_scrofulaceum"
 ## [64] "NR_116538.1_Mycobacterium_mantenii"
 ## [65] "NR_104956.1_Mycobacterium_timonense"
 ## [66] "NR_117219.1_Mycobacterium_avium"
 ## [67] "NR_179381.1_Mycobacterium_senriense"
 ## [68] "NR_116063.1_Mycobacterium_bouchedurhonense"
 ## [69] "NR_117222.1_Mycobacterium_colombiense"
 ## [70] "NR_117221.1_Mycobacterium_ariosiense"
 ## [71] "NR_116262.1_Mycobacterium_marseillense"
 ## [72] "NR_117228.1_Mycobacterium_intracellulare"
 ## [73] "NR_146658.1_Mycobacterium_paraintracellulare"
 ## [74] "NR_044840.1_Mycobacterium_intermedium"
 ## [75] "NR_135729.1_Mycobacterium_bourgelatii"
 ## [76] "NR_118879.1_Mycobacterium_gastri"
 ## [77] "NR_121712.2_Mycobacterium_kansasii"
 ## [78] "NR_179497.1_Mycobacterium_innocens"
 ## [79] "NR_156953.1_Mycobacterium_persicum"

```
## [80] "NR_179496.1_Mycobacterium_pseudokansasii"
## [81] "NR_179498.1_Mycobacterium_attenuatum"
## [82] "NR_029298.1_Mycobacterium_conspicuum"
## [83] "NR_178718.1_Mycobacterium_paraense"
## [84] "NR_042793.1_Mycobacterium_saskatchewanense"
## [85] "NR_179522.1_Mycobacterium_terraeassiliense"
## [86] "NR_104909.1_Mycobacterium_interjectum"
## [87] "NR_025000.1_Mycobacterium_kubicae"
## [88] "NR_028940.1_Mycobacterium_palustre"
## [89] "NR_125568.1_Mycobacterium_europaeum"
## [90] "NR_118110.1_Mycobacterium_parascrofulaceum"
## [91] "NR_025268.1_Mycobacterium_heidelbergense"
## [92] "NR_025225.1_Mycobacterium_parmense"
## [93] "NR_181114.1_Mycobacterium_helveticum"
## [94] "NR_042665.1_Mycobacterium_stomatepieae"
## [95] "NR_042223.1_Mycobacterium_florentinum"
## [96] "NR_029223.1_Mycobacterium_genavense"
## [97] "NR_044791.1_Mycobacterium_triplex"
## [98] "NR_028808.1_Mycobacterium_montefiorensis"
## [99] "NR_115209.1_Mycobacterium_sherrisii"
## [100] "NR_117227.1_Mycobacterium_simiae"
## [101] "NR_179523.1_Mycobacterium_rhizamassiliense"
## [102] "NR_179524.1_Mycobacterium_numidiamassiliense"
## [103] "NR_146370.1_Mycobacterium_shigaense"
## [104] "NR_041898.1_Mycobacterium_lentiflavum"
## [105] "NR_179612.1_Mycobacterium_ahvazicum"
```

```
root <- root(tree, outgroup = "NR_044590.1_Hoyosella_altamirensis")
# Define the name of the unknown sample
unknown_sample1 <- "AOKDPPHH_03798"
unknown_sample2 <- "AOKDPPHH_03803"
unknown_sample3 <- "AOKDPPHH_04179"
unknown_sample4 <- "AOKDPPHH_04184"

# Create a vector of colors for the tips
# Make the unknown sample a different color from the rest
tip_colors <- "black"
tip_colors[str_detect(tree$tip.label, unknown_sample1)] <- "red"
tip_colors[str_detect(tree$tip.label, unknown_sample2)] <- "red"
tip_colors[str_detect(tree$tip.label, unknown_sample3)] <- "red"
tip_colors[str_detect(tree$tip.label, unknown_sample4)] <- "red"

# Plot the tree
plot(tree, tip.color = tip_colors, cex = 0.7)
node.labels(tree$node.label, cex = 0.7, frame = "none", adj = c(1, -0.5))
```

Antimicrobial Resistance Gene Identification

NTMs are usually resistant to many antibiotics, thus infections are hard to treat. Using AMRFinderPLUS allows us to detect known antimicrobial resistance genes in NTM_22-11. This could help researchers understand what kinds of antibiotics are ineffective against NTM_22-11 so that they can create an effective treatment.

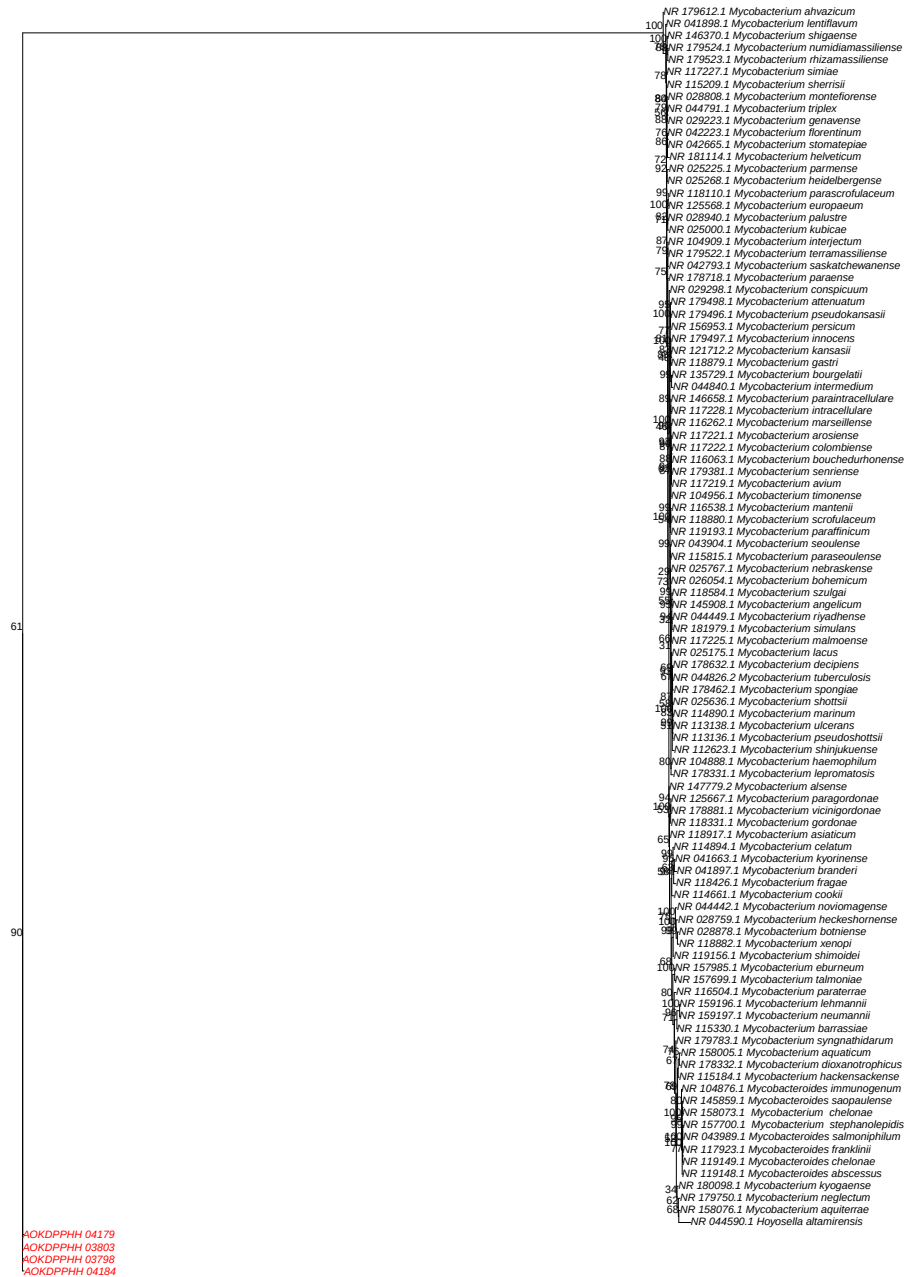


Figure 1: Phylogenetic tree of the unknown NTM sample based on the 16S rRNA gene.

```

# Running AMRFinderPlus
# Place the docker command for AMRFinderPlus here
docker pull ncbi/amr:latest

docker run --name=amrfinder \
--rm \
--volume=$PWD:/home \
--workdir=/home \
ncbi/amr:latest amrfinder \
  --threads 8 \
  -a prokka \
  -p ./prokka_out/prokka_annotation.faa \
  -g ./prokka_out/prokka_annotation.gff --print_node \
  --plus \
  --protein_output ./amr_genes/amrfinder_pro.fasta >./amr_genes/amrfinder_pro_results.tsv

# Read in the AMR results
# Extract the Protein.identifier, Gene.symbol, and Sequence.name columns into a new data frame
# Rename the columns to Protein ID, Gene Symbol, and Sequence Name
amrs <- read.csv("amr_genes/amrfinder_pro_results.tsv", sep = "\t")
amr_summary <- data.frame(amrs) %>%
select (Protein.id, Element.name, Element.symbol) %>%
rename(Protein_ID = Protein.id, Gene_Symbol = Element.symbol, Sequence_Name = Element.name)

kable(amr_summary)

```

Protein_ID	Sequence_Name	Gene_Symbol
AOKDPPHH_009666	multidrug efflux transporter outer membrane subunit EmrC	emrC
AOKDPPHH_009667	multidrug efflux MFS transporter periplasmic adaptor subunit EmrA	emrA
AOKDPPHH_009668	multidrug efflux MFS transporter permease subunit EmrB	emrB
AOKDPPHH_016255	APH(6) family putative aminoglycoside O-phosphotransferase	aph(6)
AOKDPPHH_017266	opper-binding protein CopB	copB
AOKDPPHH_017267	accase-like oxidase CopA	copA
AOKDPPHH_018751	family subclass B3 metallo-beta-lactamase	blaL1
AOKDPPHH_027333	aminoglycoside 6'-N-acetyltransferase	aac(6')
AOKDPPHH_029110	class A beta-lactamase	bla
AOKDPPHH_032556	multidrug efflux RND transporter outer membrane subunit SmeF	smeF
AOKDPPHH_035779	AD(+)-rifampin ADP-ribosyltransferase	arr

```

# amr_summary %>%
# kable("latex", booktabs = TRUE, caption = "AMR Gene Summary",
#       col.names = c("Protein ID", "Gene Symbol", "Sequence Name")) %>%
# kable_styling(latex_options = c("striped", "hold_position"))

```

Conclusions

From observing the phylogentic tree, multiple Mycobacterium speicies could be found across the branches. This means that across most Mycobacterium speicies, it is similar to the NTM_22-11 bacteria. With four unkown speicies in red, they represent multiple 16s rRna gene copies from NTM_22-11 or similar contigs from the same genome. We also observed that the phylogentic tree is compressed because the 16s rRNA

gene fits very closely to other Mycobacterium species, so it is hard to differentiate NTM_22-11 from other NTMs. Thus, there is no final conclusion on whether NTM_22-11 is a novel species. More in depth analysis such as nucleotide identification may be used to further investigate whether NTM_22-11 is a novel bacteria species. Despite not having a definitive answer, there are information on what antibiotics the potential NTM is resistant to.

Source

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5631500/>